



QUANTITATIVE SKILLS

QUESTION BANK – MID 2

Short answer questions

- 1) At what rate of interest will a sum of **45000 Rs** give **13500 Rs** simple interest in **6 years**?

$$P = 45000, \text{ si} = 13500, t=6$$

$$r = \frac{SI \times 100}{P \times t} = \frac{13500 \times 100}{45000 \times 6} = 5\%.$$

Answer: 5% p.a.

- 2) **720 Rs** is divided among three persons **A, B, C** in the ratio **2/3 : 3/4 : 4/5**. Find **B's share**.

Ratio = 2/3 : 3/4 : 4/5. Multiply by 60 → 40 : 45 : 48. Sum = 133.

$$B's \text{ share} = \frac{45}{133} \times 720 = \frac{32400}{133} \approx \mathbf{243.61Rs.}$$

(Exact: 32400/133Rs.)

- 3) Two persons **A and B** divided their property in the ratio **4 : 7**. If **B got 450000 Rs more than A**, find the **total property**.

$$\text{Ratio } 4 : 7. \text{ Difference} = 3k = 450000 \Rightarrow k = 150000. \text{ Total} = 11k = 11 \times 150000 = \mathbf{1,650,000Rs.}$$

- 4) A train running at **72 km/hr** crosses a pole in **12 seconds**. Find the **length of the train**.

$$72 \text{ km/h} = 72 \times \frac{5}{18} = 20\text{m/s. Length} = 20 \times 12 = \mathbf{240m.}$$

- 5) Two trains with speeds **65 km/hr** and **85 km/hr** start from two places at the same time. Distance between them is **1350 km**. In how many hours will they **meet**?

$$\text{Relative speed} = 65 + 85 = 150\text{km/h. Time} = 1350/150 = \mathbf{9\text{hours.}}$$

- 6) Find **compound interest** on a sum of **150000 Rs** at **6% per annum** for **2 years**.

$$A = 150000(1.06)^2 = 150000 \times 1.1236 = 168540. \text{ CI} = 168540 - 150000 = \mathbf{18,540Rs.}$$



7) Walking at **4/5 of usual speed**, a man reaches **8 minutes late**. Find the **usual time** taken.

$$\text{Usual time} = t. \text{ New time} = \frac{5}{4}t. \text{ Delay} = \frac{1}{4}t = 8\text{min} \rightarrow t = 32\text{min}.$$

Answer: 32 minutes

8) An Aeroplane covers a distance at **240 km/hr.** in **5 hours**. At what speed should it travel to cover the same distance in **7.5 hours**?

$$\text{Distance} = 240 \times 5 = 1200\text{km}. \text{ Required speed} = 1200/7.5 = \mathbf{160\text{km/h}}.$$

9) Two numbers are **20%** and **40%** more than a third number. Find the **ratio** of the two numbers.

$$\text{Numbers} = 1.2x \text{ and } 1.4x. \text{ Ratio} = 1.2:1.4 = 12:14 = \mathbf{6:7}.$$

10) A **60 L** mixture contains milk and water in the ratio **7:3**. How much **water** should be added to make it **3:2**?

$$\text{Milk} = \frac{7}{10} \times 60 = 42\text{L}, \text{ water} = 18\text{L}. \text{ After adding } w: 42:(18+w) = 3:2.$$

$$84 = 54 + 3w \Rightarrow w = 10\text{L}. \mathbf{\text{Answer: 10 L}}$$

11) Find compound interest on **100000 Rs** for **3 years** at **4% per annum**.

$$A = 100000(1.04)^3 = 100000 \times 1.124864 = 112486.40.$$

$$\text{CI} = 112486.40 - 100000 = \mathbf{12,486.40\text{Rs}}.$$

12) The prices of a **laptop** and a **mobile** are in the ratio **5:3**. If the laptop costs **18000 Rs more**, find the **price of the mobile**.

$$\text{Ratio } 5:3. \text{ Difference} = 2k = 18000 \Rightarrow k = 9000. \text{ Mobile} = 3k = \mathbf{27,000\text{Rs}}.$$

13) A **90 L** mixture contains milk and water in the ratio **5:4**. How much **water** must be added to make the ratio **4:5**?

$$\text{Milk} = \frac{5}{9} \times 90 = 50\text{L}, \text{ water} = 40\text{L}. \text{ Need } w \text{ so } 50:(40+w) = 4:5.$$

$$250 = 160 + 4w \Rightarrow 4w = 90 \Rightarrow w = 22.5\text{L}. \mathbf{\text{Answer: 22.5 L}}$$



14) A train is moving with a speed of **75 km/hr**. How many **meters** will it cover in **3 minutes**?

$$75 \text{ km/h} = 75 \times \frac{5}{18} = 20\frac{5}{6} \text{ m/s. In 3 min (180 s):}$$

$$\text{distance} = 20.8333 \dots \times 180 = 3750 \text{ m.}$$

Answer: 3750 m

15) A train of length **250 m** crosses a platform of length **350 m** in **30 seconds**. Find the **speed** of the train in km/hr.

$$\text{Total distance} = 250 + 350 = 600 \text{ m in } 30 \text{ s}$$

$$\text{speed} = 600/30 = 20 \text{ m/s} = 20 \times 3.6 = \mathbf{72 \text{ km/h.}}$$

Long answer questions

1) CI on Rs.100000 for 2 years at 10% p.a., compounded half-yearly

$$\text{Rate per half-year} = 10\%/2 = 5\%. \text{ Periods} = 2 \text{ years} \times 2 = 4.$$

$$\text{Amount} = 100000(1.05)^4 = 121550.625.$$

$$\text{Compound interest} = 121550.625 - 100000 = \mathbf{21,550.63 \text{ Rs.}}$$

2) Salaries ratio 2:3, after +5000 each → new ratio 9:13. Find A.

$$\text{Let salaries} = 2k \text{ and } 3k.$$

$$\frac{2k+5000}{3k+5000} = \frac{9}{13} \rightarrow 13(2k+5000) = 9(3k+5000)$$

$$26k + 65000 = 27k + 45000 \Rightarrow k = 20000.$$

$$\text{Initial salary of A} = 2k = \mathbf{40,000 \text{ Rs.}}$$

3) Total 61 km in 9 h; speeds 4 km/h (foot) and 9 km/h (bicycle). Find distance on foot.



Let foot distance = x . Time: $x/4 + (61 - x)/9 = 9$.
Solve: $9x + 4(61 - x) = 324 \Rightarrow 5x = 80 \Rightarrow x = 16\text{km}$.

4) Train 280 m, speed 72 km/h, platform 220 m. Time to cross platform?

$72 \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s}$. Total distance = $280 + 220 = 500 \text{ m}$.
Time = $500/20 = 25 \text{ seconds}$.

5) Coins 25p, 10p, 5p in ratio 1:2:3; total = Rs.60. Find number of 10p coins.

Let counts = $k, 2k, 3k$. Value (paise) = $25k + 10(2k) + 5(3k) = 25k + 20k + 15k = 60k \text{ paise}$.
In rupees = $60k/100 = 0.6k = 60 \Rightarrow k = 100$.
Number of 10p coins = $2k = 200$.

6) Difference between SI and CI for 2 years at 10% is Rs.600. Find sum.

Difference = Pr^2 (for 2 years at rate r per year). Here $r = 0.10$.
So $P \times (0.1)^2 = 600 \Rightarrow P \times 0.01 = 600 \Rightarrow P = 60,000 \text{ Rs}$.

7) Two numbers in ratio 3:5. After adding 12 each, ratio becomes 5:7. Find sum.

Let numbers = $3k, 5k$. $\frac{3k+12}{5k+12} = \frac{5}{7}$. Solve: $k = 6$.
Numbers = 18 and 30. Sum = 48.

8) Same coin-ratio problem with total Rs.30. How many 5p coins?

As before, value in rupees = $0.6k = 30 \Rightarrow k = 50$.
Number of 5p coins = $3k = 150$.



9) Four successive 3 km stretches at speeds 10,20,30,60 km/h. Average speed?

$$\text{Total distance} = 12\text{km. Total time} = \sum(3/\text{speed}) = 3\left(\frac{1}{10} + \frac{1}{20} + \frac{1}{30} + \frac{1}{60}\right)$$

$$\text{Inside sum} = \frac{6+3+2+1}{60} = \frac{12}{60} = \frac{1}{5}. \text{ Total time} = 3 \times \frac{1}{5} = \frac{3}{5}\text{h.}$$

$$\text{Average speed} = 12 \div \frac{3}{5} = \mathbf{20\text{km/h.}}$$

10) Boat: 16 km downstream in 2 h, upstream in 4 h. Speed in still water?

$$\text{Downstream speed} = 16/2 = 8\text{km/h. Upstream} = 16/4 = 4\text{km/h.}$$

$$\text{Boat speed in still water} = (8 + 4)/2 = \mathbf{6\text{km/h.}}$$

11) Money triples in 6 years (simple interest). In how many years becomes 25 times?

$$\text{Triple} \rightarrow 1 + 6r = 3 \Rightarrow 6r = 2 \Rightarrow r = \frac{1}{3} = 33\frac{1}{3}\% \text{p.a.}$$

$$\text{For 25 times: } 1 + rt = 25 \Rightarrow t = \frac{24}{r} = 24 \div \frac{1}{3} = \mathbf{72\text{years.}}$$

12) Boys: Girls = 7:8. Boys +20%, Girls +10%. New ratio?

$$\text{New counts} \propto 7 \times 1.2 : 8 \times 1.1 = 8.4 : 8.8 = 84 : 88 = \mathbf{21 : 22.}$$

13) Salaries 2:3. After +5000 each new ratio 5:7. Find initial A.

$$\text{Let salaries} = 2k, 3k. \frac{2k+5000}{3k+5000} = \frac{5}{7}. \text{ Solve } \rightarrow k = 10000.$$

$$\text{Initial A} = 2k = \mathbf{20,000\text{Rs.}}$$

14) Distances: 200 km @80, 300 km @60, 100 km @50. Average speed?

$$\text{Times: } 200/80 = 2.5\text{h, } 300/60 = 5\text{h, } 100/50 = 2\text{h. Total} = 600\text{km in } 9.5\text{h.}$$

$$\text{Average speed} = 600/9.5 = \frac{1200}{19} \approx \mathbf{63.1579\text{km/h}} (\approx 63.16 \text{ km/h}).$$



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15) Boat: 30 km downstream in 5 h, upstream in 6 h. Speed in still water?

Downstream = $30/5 = 6\text{km/h}$. Upstream = $30/6 = 5\text{km/h}$. Boat speed = $(6 + 5)/2 = 5.5\text{km/h}$.